

ACC250: Intro to Financial Accounting
Ch2. The Balance Sheet

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- Apply the duality of effects and the debit/credit framework.
- Run Steps 1–4 of the accounting cycle, ending with the trial balance.
- Prepare a classified balance sheet and compute the current ratio.
- Value assets under the cost principle.

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Three different types of business activities:

- ❶ Financing - The raising and repayment of capital.
 - ▶ Selling stocks to shareholders
 - ▶ Borrowing money from banks
 - ▶ Paying dividends to shareholders
- ❷ Investing - The purchase of assets that are not part of the core business.
 - ▶ Buying equipment
 - ▶ Buying land
 - ▶ Buying a new building
- ❸ Operating - The core activities of a business that generate revenue (incl. R & E).
 - ▶ Selling products
 - ▶ Buying supplies
 - ▶ Paying employees' salaries
 - ▶ Paying rent for office space

Business Activities - Classification exercise

- **F I O**: Choose F (Financing), I (Investing), or O (Operating).
- **NI**: O if the activity affects Net Income (through Revenue or Expense), otherwise X.

Business Activities	F I O	NI
Two owners of the NoodleCake decided to contribute \$5,000 each.	<u>F</u>	<u>X</u>
NoodleCake received \$20,000 cash in exchange for its promise to repay the loan in two years.	<u>F</u>	<u>X</u>
NoodleCake purchased a logo by paying \$300 cash to a designer.	<u>I</u>	<u>X</u>
NoodleCake bought \$9,600 of Equipment on credit.	<u>I</u>	<u>X</u>
NoodleCake paid \$1,000 cash to employees for salaries.	<u>O</u>	<u>(-)</u>
NoodleCake received \$2,000 cash from customers for products sold.	<u>O</u>	<u>(+)</u>

Transactions

Financial activities that involve the exchange of goods, services, or money.

- All transactions affect the basic accounting equation: $A = L + SHE$.
- Two types:
 - ▶ Operating activities: Affect NI (mainly covered in Ch3).
 - ▶ Financing and Investing activities: Do not affect NI (covered in Ch2).
- **Activities but not transactions:**
 - ▶ Promising to hire employees
 - ▶ Exchange of stocks between shareholders

A chart of accounts

A company manages its own [chart of accounts](#) to record transactions.

Examples of common accounts:

Account	Class	Definition
Cash	<u>A</u>	Money available for immediate use, including currency, coins, and balances in checking and savings accounts.
Supplies	<u>A</u>	Items used in the course of business operations that are expected to be consumed within a short period.
Equipment	<u>A</u>	Long-term assets such as machinery, computers, or vehicles used in business operations.
Logo and Trademarks	<u>A</u>	Intangible assets representing the company's brand identity and legally protected symbols or names.
Software	<u>A</u>	Computer programs and applications owned or licensed by the company for business use.
Accounts Payable	<u>L</u>	Amounts the company owes to suppliers for goods or services purchased on credit.
Notes Payable	<u>L</u>	Written promises to pay a specific amount of money at a future date, often with interest.
Common Stock	<u>SHE</u>	The basic ownership shares issued to investors, representing ownership in the company.
Retained Earnings	<u>SHE</u>	Cumulative net income that has been kept in the company rather than distributed to shareholders as dividends.

Duality of Effects

Each transaction has at least two effects on the basic accounting equation:

- $A = L + SHE$

Analyze the effects of the following transactions:

Business Activities	A	L	SHE
1. Two owners of the NoodleCake decided to contribute \$5,000 each.	+		+
2. NoodleCake received \$20,000 cash in exchange for its promise to repay the loan in two years.	±	±	
3. NoodleCake purchased a logo by paying \$300 cash to a designer.	+ / -		
4. NoodleCake bought \$9,600 of Equipment on credit.	±	±	
5. NoodleCake paid \$1,000 cash to employees for salaries.	-		-
6. NoodleCake received \$2,000 cash from customers for products sold.	±		±

Debit/Credit Framework: 1. What are Debit and Credit?

Debit(DR): Left

Credit (CR): Right

Debit/Credit Framework: 2. What are Debit and Credit Accounts?

$$A = L + SHE$$

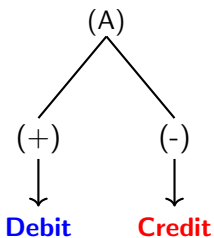


Those on
the **left-hand** side:
Debit Accounts

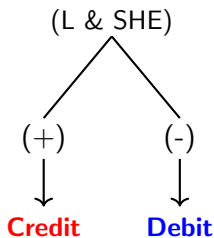


Those on
the **right-hand** side:
Credit Accounts

Debit accounts



Credit accounts



- When Debit accounts increase: Debit.
- When Debit accounts decrease: Credit.
- When Credit accounts increase: Credit.
- When Credit accounts decrease: Debit.

Debit/Credit Framework

	A	L	SHE
Debit / Credit account	<u>Debit acc.</u>	<u>Credit acc.</u>	<u>Credit acc.</u>
Increase	<u>D (L)</u>	<u>C (R)</u>	<u>C (R)</u>
Decrease	<u>C (R)</u>	<u>D (L)</u>	<u>D (L)</u>

EX. Prepare journal entries for the following transactions under the debit/credit framework:

- ① Two owners of NoodleCake each contributed \$5K.

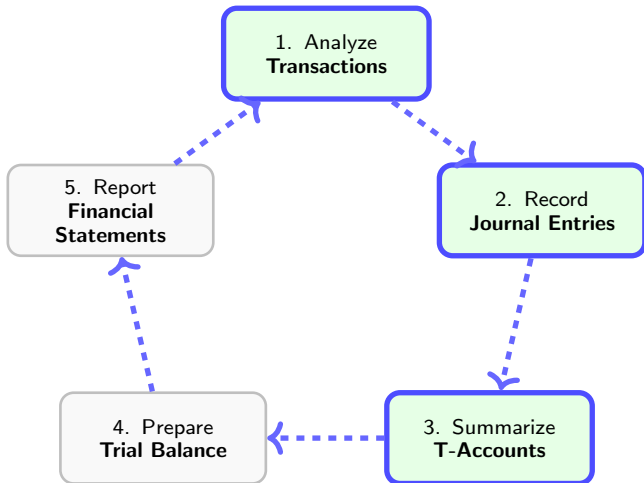
Cash (+A)	10K
Common Stock (+SHE)	10K

- ② Noodlecake pays \$300 cash to create the company's logo.

Logo and Trademarks (+A)	300
Cash (-A)	300

Accounting Cycle

A systematic accounting process is used to capture and report the financial effects of a company's transactions.



Note: Steps 1–3 are covered in this chapter.

Step 1. Analyze Transactions

Questions for Every Transaction

- 1 Is it a transaction?
- 2 Which accounts are affected?
- 3 How are they affected? (increase or decrease)

Example: You bought \$500 worth of supplies and paid cash.

- Transaction? It affects the basic accounting equation, so it's a transaction.
- What accounts? Supplies (Asset) and Cash (Asset) are affected.
- How affected? Supplies increases, Cash decreases.

A	= L	+ SHE
Supplies +500		
Cash -500		

Step 1. Analyze Transactions - Examples

- ① Two owners of the NoodleCake decided to contribute \$5K each.

A	= L	+ SHE
<u>Cash +10,000</u>		<u>Common Stock +10,000</u>

- ② NoodleCake received \$20K cash in exchange for its promise to repay the loan in two years.

A	= L	+ SHE
<u>Cash +20,000</u>	<u>Notes Payable +20,000</u>	

- ③ NoodleCake bought \$9,600 of Equipment on credit.

A	= L	+ SHE
<u>Equipment +9,600</u>	<u>Accounts Payable +9,600</u>	

Step 2. Record Journal Entries

EX. Consider the following transactions. Prepare journal entries for each:

- ① Two owners of NoodleCake each contributed \$5K.

Cash (+A)	10K
Common Stock (+SHE)	10K

- ② Noodlecake pays \$300 cash to create the company's logo.

Logo and Trademarks (+A)	300
Cash (-A)	300

- ③ NoodleCake received \$20K cash in exchange for a promise to repay the loan in 2 years.

Cash (+A)	20K
Notes Payable (+L)	20K

- ④ Noodlecake purchases and receives \$9,600 in equipment (e.g., computers), in exchange for its promise to pay \$9,600 at the end of the month.

Equipment (+A)	9,600
Accounts Payable (+L)	9,600

- 5 Noodlecake pays \$5K to the equipment supplier in (d).

Accounts Payable (-L)	5K
Cash (-A)	5K

- 6 Noodlecake signs a contract with a programmer for program code for the Enchanted World game app for \$9K. No code has been received yet.

No journal entries required.

- 7 Noodlecake receives the \$9K of app game code ordered in (f), pays \$4K cash, and promises to pay the remaining \$5K next month.

Software (+A)	9K
Cash (-A)	4K
Accounts Payable (+L)	5K

- 8 Noodlecake receives supplies costing \$600 on account.

Supplies (+A)	600
Accounts Payable (+L)	600

Step 3. Summarize T-Accounts

Ledger Accounts, T-Accounts, and General Ledger

- Transactions are posted to (i.e., summarized by) (Ledger Accounts).
- A complete list of Ledger Accounts is called General Ledger.
- T-account is a visual representation of the (Ledger Accounts).
- We use T-accounts in this course.

Cash	
Beg. 0	
(a) <u>10,000</u>	<u>300</u> (b)
(c) <u>20,000</u>	<u>5,000</u> (e)
	<u>4,000</u> (g)
End. <u>20,700</u>	

Step 3. Summarize T-Accounts

Balances of T-accounts

- Debit accounts have normally debit balances.
- Credit accounts have normally credit balances.

How to get transactions (in journal entries) posted to T-accounts?

- Get the beginning balance of the T-account from the previous period.
- For each account, go through transactions that affect the account.
 - ▶ For those with debit values, add the debit value to the T-account.
 - ▶ For those with credit values, add the credit value to the T-account.
- Get the ending balance of the T-account.

<u>Asset Accounts</u>			<u>Liab. Accounts</u>			<u>SHE. Accounts</u>	
Beg. Bal.			Beg. Bal.			Beg. Bal.	
		=			+		
End. Bal.			End. Bal.			End. Bal.	

Step 3. Summarize T-Accounts

ASSETS

Cash

Beg. 0	
(a) 10,000	300 (b)
(c) 20,000	5,000 (e)
	4,000 (g)
End. 20,700	

Supplies

Beg. 0	
(h) 600	
End. 600	

Equipment

Beg. 0	
(d) 9,600	
End. 9,600	

Logo & Trademarks

Beg. 0	
(b) 300	
End. 300	

Software

Beg. 0	
(g) 9,000	
End. 9,000	

LIABILITIES

Accounts Payable

	Beg. 0
(e) 5,000	9,600 (d)
	5,000 (g)
	600 (h)
	End. 10,200

Notes Payable

	Beg. 0
	20,000 (c)
	End. 20,000

SHAREHOLDERS' EQUITY

Common Stock

	Beg. 0
	10,000 (a)
	End. 10,000

Step 4. Prepare Trial Balance

- Take all the debits and credits from the T-Accounts
- Put them in the Trial Balance.
- Calculate the total debits and credits.
- Check if the **total debits** equal the **total credits**.
- **If it does not, there is an error in the journal entries.**

Account Title	Debit (\$)	Credit (\$)
Cash	<u>20,700</u>	
Supplies	<u>600</u>	
Equipment	<u>9,600</u>	
Logo & Trademarks	<u>300</u>	
Software	<u>9,000</u>	
Accounts Payable		<u>10,200</u>
Notes Payable		<u>20,000</u>
Common Stock		<u>10,000</u>
Total	<u>40,200</u>	<u>40,200</u>

Step 4. Prepare Trial Balance

- If the total debits do not equal the total credits, there's an error in the journal entries.
- If the total debits equal the total credits, the journal entries are likely correct.
- Having the same balances does not guarantee the journal entries are correct.

- **Assets** are grouped into **current assets** and **non-current assets**.
 - ▶ **Current assets**: those expected to be converted to cash or used up within 1 year (e.g., Cash, Supplies).
 - ▶ **Non-current assets**: those expected to provide benefits for more than 1 year (e.g., Equipment, Logo & Trademarks, Software).
- **Liabilities** are grouped into **current liabilities** and **non-current liabilities**.
 - ▶ **Current liabilities**: those expected to be paid within 1 year (e.g., Accounts Payable, Notes Payable).
 - ▶ **Non-current liabilities**: those expected to be paid more than 1 year from now (e.g., Notes Payable).

Classified Balance Sheet

NoodleCake Studio, Inc.
Balance Sheet
As of August 31, 20XX

Assets	
Current Assets	
Cash	\$20,700
Supplies	600
Total Current Assets	\$21,300
Equipment	9,600
Logo & Trademarks	300
Software	9,000
Total Assets	<u>\$40,200</u>
Liabilities and Shareholders' Equity	
Current Liabilities	
Accounts Payable	\$10,200
Total Current Liabilities	\$10,200
Notes Payable	20,000
Total Liabilities	<u>\$30,200</u>
Stockholders' Equity	
Common Stock	\$10,000
Retained Earnings	\$0
Total Stockholders' Equity	<u>\$10,000</u>
Total Liabilities and Shareholders' Equity	<u>\$40,200</u>

Current Ratio

Current Ratio

A measure of a company's ability to pay its short-term obligations.

Calculated as: $\frac{\text{Current Assets}}{\text{Current Liabilities}}$

- Current Ratio > 1: the company is able to pay its current liabilities.
- Current Ratio < 1: the company is **not** able to pay its current liabilities.

EX. From the previous example:

- Current assets: \$21,300
- Current liabilities: \$10,200
- Current Ratio: $\frac{\$21,300}{\$10,200} = 2.09$
- Implications: The company is able to pay its current liabilities (to suppliers, employees, etc.) using its current assets (e.g., Cash, Accounts Receivable).

Cost Principle

Assets are initially recorded at their **acquisition cost**, which includes all costs necessary to acquire the asset and prepare it for its intended use.

- Asset values are initially determined by the cost principle.
(i.e., how much the company paid to acquire the assets.)
- **Not** the current market value of the assets.
- Subsequently, the value is adjusted for depreciation and other factors.
(will be covered later.)

EX. Think about this:

- The firm paid \$9,600 to acquire the equipment.
 - ▶ Historical acquisition cost: \$9,600 (following the cost principle).
- To buy the same one today, the firm would need to pay \$10K.
 - ▶ Current market value: \$10K.
- Historical acquisition cost is **not** the current market value of the asset.
- In B/S, the asset value is the historical acquisition cost.